MARTIN ANDREEV ASENOV

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EDUCATION

The University of Edinburgh and Heriot-Watt University Sep 2017 - Jan 2022
Ph.D. in Robotics and Autonomous Systems

- "Architectures for Online Simulation-Based Inference Applied to Robot Motion Planning"
- Supervised by Prof. Subramanian Ramamoorthy and Dr. Kartic Subr

The University of Edinburgh and Heriot-Watt University Sep 2016 - Aug 2017
MRes in Robotics and Autonomous Systems

- "Active Sensing of Spatiotemporal Phenomena with a UAV"
- Supervised by Prof. Subramanian Ramamoorthy

The University of Edinburgh Sep 2012 - May 2016
BSc (Hons) Artificial Intelligence and Computer Science

- "Dynamic Model of Interactions Between Orientation Selective Neurons in Primary Visual Cortex"
- Supervised by Dr. Mark van Rossum
- International Exchange Student at The University of California, Irvine (Sep 2014 - Jun 2015)

WORK EXPERIENCE

Parexel AI Labs Sep 2025 - Current
Senior Machine Learning Engineer I *Edinburgh, UK (Remote)*

- Agentic workflows and automated document-processing libraries, used internally by 20K+ employees

Huawei Technologies Research & Development Aug 2022 - Sep 2025
Senior Machine Learning Engineer *Edinburgh, UK*

- Research in online learning and transformer-based foundational models for timeseries predictions
- Optimization of LLMs and vision models placement on different types of hardware
- Lead a team of seven people developing methods for runtime estimation of neural networks
- Lead development of a production timeseries framework for a serverless platform resource predictions

Hyperscience Apr 2022 - Jul 2022
Machine Learning Engineer *Edinburgh, UK (Remote)*

- Improved accuracy of the model with domain adaptation and reducing preprocessing steps
- Achieved MVP with 15% improvement in accuracy within 2 months of starting

The University of Edinburgh Apr 2021 - Mar 2022
Research Associate *Edinburgh, UK*

- Breast cancer classification based on combined pCLE images and Raman spectroscopy
- Adaptive scanning strategies for ultrasound examinations

Amazon, PrimeAir Oct 2018 - Dec 2018
Applied Scientist Intern *Graz, Austria*

- Created a new pipeline for video segmentation of people, animals and objects
- Implemented data augmentation with gradient reversal layer to further improve accuracy

The University of Edinburgh Jan 2017 - Jun 2017; Nov 2017 - Sep 2018
Research Assistant *Edinburgh, UK*

- Collaboration between DSTL, Heriot-Watt University, BMT and REDALERT
- Data-driven fluid simulation approach to characterize gas dynamics

Thales Avionics Jul 2015 - Aug 2015
Software Integration and Characterization Developer *Irvine, California, USA*

- Extended Google's Blockly visual programming editor to a test automation tool
- Assisted in training a support vector machine, able to estimate performance tests without running them

SELECTED PUBLICATIONS

Retrieval or Representation? Reassessing Benchmark Gaps in Multilingual and Visually Rich RAG, M. Asenov, K. Benkirane, D. Goldwater, A. Ghodsi, I Can't Believe It's Not Better: Where Large Language Models Need to Improve, **ICLR '26**

DISCO: Document Intelligence Suite for COmparative Evaluation, K. Benkirane, D. Goldwater, M. Asenov, A. Ghodsi, Multimodal Intelligence: Next Token Prediction & Beyond, **ICLR '26**

Higher Acceptance Rates for Speculative Decoding with Randomised Drafting, W. Toner, M. Asenov, R. Singh, A. Joosen, Tiny Titans: The next wave of On-Device Learning for Foundational Models (TTODLer-FM), **ICML '25**

Lightweight Online Adaption for Time Series Foundation Model Forecasts, T. L. Lee, W. Toner, R. Singh, A. Joosen, M. Asenov, **ICML '25**

Serverless Cold Starts and Where to Find Them, A. Joosen, A. Hassan, M. Asenov, R. Singh, L. Darlow, J. Wang, Q. Deng, A. Barker, **EuroSys '25**

Performance of Zero-Shot Time Series Foundation Models on Cloud Data, W. Toner, T. L. Lee, A. Joosen, R. Singh, M. Asenov, I Can't Believe It's Not Better: Challenges in Applied Deep Learning Workshop, **ICLR '25**

DAM: Towards a Foundation Model for Forecasting, L. N. Darlow, Q. Deng, A. Hassan, M. Asenov, R. Singh, A. Joosen, A. Barker, A. Storkey, **ICLR '24**

FoldFormer: Sequence Folding and Seasonal Attention for Fine-grained Long-term FaaS Forecasting, L. N. Darlow, A. Joosen, M. Asenov, Q. Deng, J. Wang, A. Barker, **EuroMLSys '23**

Learning to Score: Tuning Cluster Schedulers through Reinforcement Learning, M. Asenov, Q. Deng, G. Yeung, A. Barker, **IC2E '23**

Architectures for online simulation-based inference applied to robot motion planning, M. Asenov, **PhD Thesis**

Vision-Based System Identification and 3D Keypoint Discovery using Dynamics Constraints, M. Jaques, M. Asenov, M. Burke, T. Hospedales, Workshop on Physical Reasoning and Inductive Biases for the Real World, **NeurIPS '21**

Inversion of Ultrafast X-ray Scattering with Dynamics Constraints, M. Asenov, N. Zotev, S. Ramamoorthy, A. Kirrander, Workshop on Machine Learning and the Physical Sciences, **NeurIPS '20**

Vid2Param: Modelling of Dynamics Parameters from Video, M. Asenov, M. Burke, D. Angelov, T. Davchev, K. Subr, S. Ramamoorthy, **RAL**, Vol 5(2): 414-421. Presented at **ICRA '20**

Active Localization of Gas Leaks using Fluid Simulation, M. Asenov, M. Rutkauskas, D.T. Reid, K. Subr, S. Ramamoorthy, **RAL**, Vol 4(2): 1776-1783. Presented at **ICRA '19**